SECRETS OF THE

OPTIMIZED BRAIN

3rd EDITION

NOOTROPICS EXPERT

David Tomen

92 NOOTROPICS TO UNLOCK YOUR TRUE BRAIN POTENTIAL

$28

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Introduction

This is a comprehensive list of the most popular *nootropics* in use today. Here you'll learn what the *nootropic* is, what it does and suggested dosages.

If you are new to the world of *nootropics*, you may be wondering the best way to use a particular compound. You'll find that in this guide.

If you are an experienced neurohacker this is a great quick reference guide.

You'll find a brief description of benefits of each *nootropic*. Its specific mechanism of action. How safe it is. Suggestions on pairing it with other *nootropics* in your stack. And dosage recommendations.

You'll also come across some phrases or words that are live links. Those links will take you to an explanation of that word or phrase in the Nootropics Glossary on NootropicsExpert.com.

You'll notice that each *nootropic* listed also ends with a link. If you are reading this guide on your computer, tablet, or phone, clicking on the link will take you to a comprehensive full review of that nootropic on NootropicsExpert.com.

(Note that you can also go to NootropicsExpert.com and search for that nootropic. You'll find search results linking you to that page on the website).

Each extended review on NootropicsExpert.com includes what the *nootropic* is, and where it comes from. Including a history of that compound. You'll also find information on its *mechanism of action* in your brain. What goes wrong in your brain and how that nootropic can help fix and often correct the issue.

Each extended article also includes how the nootropic supplement *feels* when you take. Detailed dosage recommendations. Side effects, drug interactions, and the different forms available. You'll also have access to tons of scientific research and clinical studies for each nootropic.

If you enjoy this guide, consider getting a copy of my book – ***Head First*** – *The Complete Guide to Healing and Optimizing Your Brain with Nootropic Supplements*.

Head First is nearly 600-pages and *the first authoritative reference guide on nootropics written in the last 20 years*. You can see a full description of what's in *Head First* by clicking here → ***Head First***

Now a ***word of caution*** *before we get started* – always start off with the lowest effective dose of any supplement. Each of our bodies is different so you need to find out the effects of each nootropic in your body. And how it will benefit you.

This list of Nootropics is in alphabetical order. To quickly go to that listing just click on the name in the ***Table of Contents*** below.

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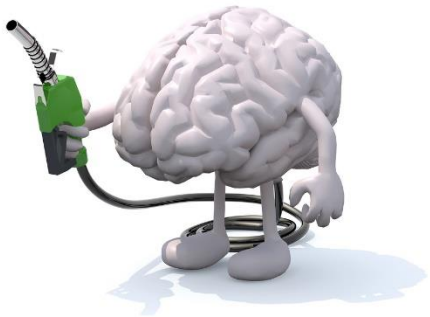
Acetyl L-Carnitine (ALCAR)

Acetyl L-Carnitine ((*R*)-3-Acetyloxy-4-trimethylammonio-butanoate) is an acetylated form of *L-Carnitine*. A derivative of the amino acid *lysine* which is naturally produced in your liver and kidneys. This synthesized form can easily cross the blood-brain barrier.

ALCAR is a nootropic because it targets your brain metabolism, boosts mitochondrial energy, and acts as a neuroprotectant. *Acetyl L-Carnitine* is a cofactor in the synthesis of *acetylcholine* which improves brain cell signaling. And makes it a powerful compliment to boost the effectiveness of other nootropics.



Acetyl-L-Carnitine boosts *memory*, mental *alertness*, fluid thought, and is a strong *antioxidant*.

The usual suggested dosage for ALCAR is 500 – 1,500 mg per day.

Learn more about Acetyl-L-Carnitine

Alpha-GPC

Alpha GPC (alpha-glycerylphosphorylcholine) is found naturally in your brain. It is a precursor to the synthesis of acetylcholine – an essential neurotransmitter involved in *memory, cognition, sleep* and *muscular control*.

Alpha GPC is more *bioavailable* than other sources of *choline* for brain benefits. Unlike *choline citrate* or *choline bitartrate*, it easily crosses the *blood-brain barrier*. This makes *Alpha GPC* a preferred choline source with experienced nootropic users.

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You naturally get *Alpha GPC* from raw beef liver, cod fish, spinach, milk, soy, and eggs. As a supplement, *Alpha GPC* derived from soy or *sunflower lecithin* is a more efficient way of boosting *acetylcholine* in your brain.

Recommended daily dosage for *Alpha GPC* is 600 mg.

Learn more about Alpha GPC

Alpha-Lipoic Acid (ALA)

Alpha-Lipoic Acid (ALA) is a sulfur-containing *fatty acid* naturally found in your body. *Lipoic acid* is unique among other *antioxidants* because it is both *water*- and *fat-soluble*.

Lipoic acid boosts the production of *acetylcholine* and glucose uptake in the brain.

Alpha Lipoic Acid has a powerful ability to neutralize the damaging effects of *oxidative stress*. Studies show *lipoic acid* even *reducing brain damage after a stroke*.

ALA regenerates other depleted antioxidants (*Vitamins C & E,* and *glutathione*), and recycles *CoQ10*. Reducing *inflammation* and getting rid of *heavy metals*. Boosting cellular energy and memory.

Recommended dosage for *Alpha-Lipoic Acid* is 200 – 400 mg per day.

Learn more about Alpha-Lipoic Acid

Aniracetam

Aniracetam (1-(4-methoxybenzoyl)-2-pyrrolidinone) is an N-side chain derivative of *Piracetam*. Thought to be 5 to 8-times more potent than *piracetam, it's fat-soluble* and has a shorter *half-life* compared to other racetams.

A potent *nootropic*, *Aniracetam* releases 200 – 300% more *acetylcholine* in the brain. Which can increase *focus*, and *clarity of thought*.

Aniracetam also reduces anxiety and depression with no sedative effects. It seems to do this by activating the *D_2 and D_3 dopamine receptors* in the brain.

Recommended dosage of *Aniracetam* is 750 – 1,500 mg per day. Do half your daily dose twice per day. And must be used with a choline supplement like *Alpha GPC* to experience its full benefit.

Learn more about Aniracetam

Ashwagandha

Ashwagandha has been used in Ayurvedic medicine for several thousand years. In Sanskrit it literally means "the smell of a horse" which implies this herb provides the vigor and strength of a stallion.

Ashwagandha extract inhibits *acetylcholinesterase*. The *enzyme* responsible for breaking down the key neurotransmitter *acetylcholine*. Boosting *memory, learning* and *cognition*.

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Ashwagandha helps regenerate *axons* and *dendrites* of brain nerve cells. And helps reconstruct *synapses*, the junctions where nerve cells communicate with other cells.

And as an *adaptogen*, *Ashwagandha* has remarkable antidepressant qualities. Shown in clinical studies to be *as effective as several prescription antidepressant and anti-anxiety drugs*.

The typical recommended dose of *Ashwagandha* is 600 – 1,000 mg twice daily.

Learn more about Ashwagandha

Artichoke Extract (Luteolin)

Artichoke Extract is a *PDE4* inhibitor. PDE4 is an enzyme that breaks down *cAMP* molecules (messenger systems that relay signals in the brain). Stopping *PDE4* then prevents the breakdown of *cAMP*.

Artichoke Extract helps generate growth of *dendrites* on neurons. And helps reconstruct *synapses*, the junctions where nerve cells communicate with other *neurons*. Boosting *memory* and restoring neural networks.

And it helps reduce inflammation in the brain by preventing the over-production of pro-inflammatory cytokines. Preventing memory loss and neurological dysfunction.

Artichoke Extract is even more powerful when combined with *Forskolin* to significantly boost *cAMP* (brain signaling) levels.

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Recommended dosage of *Artichoke Extract (standardized to 5% Cynarins)* is 900 mg per day.

Learn more about Artichoke Extract

Bacopa Monnieri

Many consider ***Bacopa Monnieri*** to be *the best nootropic* available today. The nootropic benefit of *Bacopa Monnieri* were first revealed in ancient Ayurvedic texts. In Ayurveda it's used to help memorize long passages of text. And enhance cognition.

The ancient Hindis liked it so much they named it "*Brahmi*", after the supreme god *Brahma*.

Bacopa Monnieri is an adaptogen. It helps prevent the chemical and physical effects of stress. Instead of just suppressing the symptoms like many modern antidepressants.

Bacopa Monnieri is used to reduce *anxiety*, *depression*, and *stress*. It'll boost *memory*, *concentration*, and *reaction time*. And is used for *neuroprotection*, and to balance neurotransmitters.

The compound *Bacoside A* in *Bacopa Monnieri* easily crosses the blood-brain barrier. And binds to receptor sites to increase cognition and *memory*. It has been shown effective in treating *ADHD*.

Dosage recommendation for *Bacopa Monnieri* with 45% bacosides is 450 mg.

Learn more about Bacopa Monnieri

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Berberine

Berberine is a bright yellow alkaloid extracted from plants such as Indian Barberry (tree turmeric), Oregon Grape and goldenseal.

Plants containing *Berberine* have been used in traditional *Ayurvedic* and *traditional Chinese medicine* as an *antibacterial, anti-inflammatory, anti-diarrheal, anti-microbial, anti-protozoal* and *immune-enhancing* therapy for thousands of years.

Berberine is one of the very few nootropic supplements known to activate *adenosine monophosphate-activated protein kinase (AMPK)* (another is *Resveratrol*).

The enzyme *AMPK* is the *master metabolic switch* in each of your cells. When *AMPK* levels increase you get an *energy boost, blood glucose is regulated, insulin sensitivity improves, reduced triglycerides*, and *increased fat-burning in mitochondria*.

Berberine is one of the most powerful (literally) nootropic supplements on our List of Nootropics. Not only can *Berberine* help control diabetes, but you may find *sugar cravings disappear*. No crashes late afternoon. Energy levels and stamina increase.

Dosing *Berberine* as a nootropic is 500 mg 3-times per day. *If you are dealing with a medical condition, or are taking any medication (including antibiotics), you should speak with your doctor before taking Berberine*. Especially if you're currently using blood-sugar lowering medication.

Learn more about Berberine

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Black Seed Oil

Black Seed Oil *(Nigella sativa)* is native to the Middle East, Southern Europe, and North Africa. And has been used for millennia for a variety of illnesses.

The compound *thymohydroquinone* in *Black Seed Oil* is one of the most potent *acetylcholinesterase (AChE)* inhibitors on the planet. By increasing *acetylcholine*, *AChE inhibitors* have been found to be helpful in treating *apathy, autism, dementia, Alzheimer's, schizophrenia*, and *Parkinson's Disease*.

Black Seed Oil increases *L-Tryptophan* which boosts *serotonin* levels in the brain. Providing an *antidepressant* effect following repeated dosing.

And it has been shown in animal and human studies to support *neurogenesis* and *long-term potentiation*. Showing positive effects on *learning* and *memory*. Particularly *working memory*.

Recommended dosage of *Black Seed Oil* is 1 – 3 teaspoons per day.

Learn more about Black Seed Oil

Cacao

Cacao comes from the bean of the *Theobroma cacao tree (Cocao tree)* native to Central and South America. *Cocao powder* is at the base of all things made of *chocolate*.

It stimulates the release of *phenylethylamine (PEA)* which boosts *focus* and *awareness*. And *anandamide* (the bliss molecule) which increases *happiness* levels.

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Cacao is a source of *tryptophan* which is the precursor to *serotonin*. And *theobromine* which boosts *blood flow*, is a *stimulant*, and may account for chocolate's *aphrodisiac* qualities.

Studies show that consumption of a *flavanol*-rich *cocao* drink increases *blood flow* for up to 2 hours in certain regions of the brain. Improving *cognition, memory*, and preventing *neurodegeneration*.

And *cacao* beans are one of the richest sources of *antioxidants* available. Reducing *free radicals* that would normally cause *oxidative* damage in your brain and body.

But *cacao* and *chocolate* are NOT the same thing. How *cacao* is processed makes all the difference in retaining its nootropic benefits.

Recommended dose of *Cacao flavanols* for cognitive benefit is 500 – 1,000 mg per day.

Learn more about Cacao

Caffeine

Caffeine is the most widely used *psychoactive drug* in the world. As a *nootropic*, *caffeine* helps improve *reaction time, alertness, memory*, and *mood*.

Caffeine is an *adenosine* antagonist which influences *acetylcholine*, *epinephrine (adrenaline), serotonin* and boosts the use of *dopamine*. Providing the *stimulant* effect experienced when consuming *caffeine*.

Caffeine provides a protective effect by boosting gene expression of *brain-derived neurotrophic factor (BDNF)*. Studies show chronic *caffeine* consumption may *protect against developing neurodegenerative diseases like Alzheimer's and Parkinson's*.

Caffeine improves mood within an hour of consumption. Because it increases the density of *GABA* receptors, potentiates dopamine, and causes some serotonin receptors to be more responsive.

See the full review for dosage instructions, how to deal with *tolerance*, and the benefits of using *caffeine* stacked with *L-Theanine*.

Learn more about Caffeine

Cat's Claw

Cat's Claw is a South American vine used as a nootropic with potent anti-inflammatory and antioxidant effects that support DNA repair, immune function and normal cell division.

For cognitive health, *Cat's Claw (Uncaria tomentosa)* has been used to prevent inflammation, promote cerebral circulation, fight amyloid plaques in Alzheimer's and possibly even boost acetylcholine (ACh).

The recommended daily dose for *Cat's Claw extract (Uncaria tomentosa)* is 250 to 350 mg daily.

Learn more about Cat's Claw

CBD Oil

Cannabidiol (CBD) is extracted from the *Cannabis Sativa L.* plant constituting around 40% of the plant's active compounds.

CBD does not have the same psychotomimetic (mind-altering) effects as the main plant compound *THC (Δ9-tetrahydrocannabinol)*. In fact, CBD is able to counter the mind-altering effects caused by *THC*.

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CBD Oil is used as a nootropic for its *anti-anxiety* and *antidepressant* effects, helps *alleviate pain*, reduces *oxidative stress* and *inflammation* and is *anti-tumor*.

Research into *CBD* has rapidly gained traction in the last few years. And shows potential in the treatment of *depression, neuroinflammation, epilepsy, oxidative stress, vomiting and nausea, anxiety, schizophrenia*, and other neurodegenerative problems.

Note that *CBD Oil* and *hemp oil* are not the same and should not be confused. You will not experience the same nootropic benefits with hemp oil as you will with *CBD Oil*.

Recommended starting dosage of *CBD* is 4 – 45 mg per day. Dosage of depends on the condition you are treating, body weight, and how your body reacts to *Cannabidiol*. Dosages vary from person to person.

Learn more about CBD Oil

Choline

Choline is an essential nutrient needed by your body. And cannot make it on its own. *Choline* is arguably the most basic of nootropics.

It is a water-soluble nutrient required for the health of cell membranes. *Choline* is also the precursor to the neurotransmitter *acetylcholine*. In other words, it's the step before producing *acetylcholine* in your brain.

Food sources of *choline* are egg yolks, liver, milk and other dairy products, certain grains like quinoa and amaranth, bacon, edamame, and cruciferous vegetables.

Brain health is compromised by too little *choline*. And with our modern diet it is *nearly impossible to get enough choline without supplementation*.

Common signs of *choline* deficiency can include headaches, fatigue, memory problems, and muscle pain.

The way your body uses different sources of *choline* varies depending on the source, and mechanism of action. Common *choline* sources used as nootropics are explained next.

Choline Bitartrate

Choline Bitartrate is a "choline salt", and one of the least costly sources of choline as a nootropic supplement. It's a weaker source of choline than Alpha GPC and CDP-Choline. Meaning you need to use more to achieve similar nootropic effects.

Dosing for *choline bitartrate* ranges from 500 mg – 5 grams daily.

Learn more about Choline Bitartrate

Choline Citrate

Like other sources of choline, ***choline citrate*** acts as a precursor to acetylcholine in the brain. This is a slightly more concentrated form of *choline*. It's a combination of *choline* and a derivative of *citric acid*.

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Choline Citrate is effective as a nootropic when used in large enough doses. It assists in boosting memory and recall, cognitive abilities, helps concentration and even used to treat brain diseases such as Alzheimer's.

Athletes like *Choline Citrate* for its ability to assist in cellular and muscle health and repair.

Choline Citrate is available in powder, capsule, or pill form. And you'll find it as a main ingredient in many pre-formulated nootropic stacks. Since its water soluble you can take it with water or mixed with your favorite juice.

Choline Citrate dosage can be from 500 mg to 3 grams per day.

Learn more about Choline Citrate

Citicoline (see CDP-Choline)

CDP-Choline

CDP-Choline (Cytidine Diphosphate Choline) is also known as ***Citicoline***. *CDP-choline* is a highly *bioavailable* source of *choline* that easily crosses the *blood-brain barrier*.

CDP-Choline is metabolized in the body to form *choline* and *cytidine*.[i] In the body, *choline* aids in the synthesis of acetylcholine. *Acetylcholine* is a *neurotransmitter* associated with *memory* and *learning*.

The *cytidine* in *CDP-Choline* converts to *uridine* in the body. This nucleotide is important to neural membrane synthesis which helps cognition or thinking.

CDP-Choline is an effective nootropic on its own. And works synergistically to boost the effectiveness of other nootropics. Particularly those in the *racetam family*.

Dosing of *CDP-Choline* is 250 – 750 mg per day.

Learn more about CDP-Choline

Centrophenoxine

Centrophenoxine is a water-soluble derivative of DMAE. It is a precursor to acetylcholine in the brain. And once metabolized by your liver, it readily crosses the blood-brain barrier.

DMAE is a natural chemical found in your body. It works on boosting cognition and mood. *Centrophenoxine* seems to be more effective than *DMAE* when used as a supplement.

Marketed in Europe as "Lucidril", its prescribed to treat Alzheimer's and ADHD.

Centrophenoxine is an excellent source of *acetylcholine*. You can boost the effectiveness of certain nootropics like *Piracetam* when combined with *Centrophenoxine*.

A *Centrophenoxine* dosage range of 500 – 1000 mg is considered safe.

Learn more about Centrophenoxine

Coconut & MCT Oil

Coconut oil is extracted from the seed or fruit of the *coconut palm (Cocos nucifera)*. *Unrefined coconut oil* is a potent *antioxidant, anti-tumor, anti-bacterial, anti-viral*, and *anti-fungal*.

Coconut oil is 90% saturated fat. Nearly 60% of the fats are *medium-chain fatty acids*. Including *caprylic acid C-8 (8%), capric acid, C-10 (7%)*, and lauric acid C-12 (49%).

Medium-chain fatty acids are also known as ***medium-chain triglycerides (MCTs)***. *MCTs* are produced by hydrolyzing *coconut oil* and esterifying the *fatty acids* shorter than *lauric acid (C-12)* with glycerol.

Both *coconut* and *MCT oil* provide some powerful nootropic benefits. The *MCTs* in *coconut oil* bypass your normal digestive system and go directly to your liver where they are converted to ***ketones***.

Ketones are carried throughout your body including your brain for use in the *citric acid cycle* to produce *Adenosine Triphosphate (ATP)*. *ATP* fuels your *mitochondria* which improves alertness, cognition, memory and mood.

MCTs help increase antioxidant levels in your brain. And increase *serotonin* which provides an *anti-stress* effect.

And *coconut oil* decreases *mitochondrial* dysfunction that can be caused by *Amyloid-β plaques* which are implicated in Alzheimer's disease. And drug-resistant *epilepsy* patients have found that the *ketones* provided by *coconut oil* can help reduce *seizure frequency*.

Coconut & MCT oil are a helpful addition to any nootropic stack which contains *fat-soluble* supplements. Improving the *bioavailability* of those nootropics by delivering them directly into brain cells.

Recommended dosage of *Coconut & MCT oil* is 1 tablespoon 3-times per day. Or each time you take your nootropic dose.

Learn more about Coconut & MCT Oil

Coluracetam

One of the newer racetams on the market, ***coluracetam*** works as a choline uptake enhancer.

As a nootropic, *coluracetam* enhances *concentration* and cognition, boosts *memory*, *mood* and *focus*, and treats *anxiety* disorders. It also seems to have a *stimulant* effect which improves *motivation*. Users also report *richer and fuller sound and colors*.

General dosage is 20 – 80 mg per day but depends on your reason for using *coluracetam*. And your past experience in using *racetams*.

You may want start with a lower dose and increase over time based on your experience with this nootropic.

Learn more about Coluracetam

CoQ10 & Ubiquinol

Coenzyme Q10 is naturally obtained from food such as beef, liver, sardines, mackerel, butter, and extra virgin olive oil. In an ideal world our bodies would convert enough *CoQ10* to *Ubiquinol* from our food.

But a more convenient way is giving our bodies *CoQ10* directly by using *Ubiquinol* as a supplement. No conversion necessary.

The original supplement form of *CoQ10* was *Ubiquinone*. It was updated with a newer synthesis of *CoQ10* called *Ubiquinol* which *boosts bioavailability from 40% to 90%*.

CoQ10 is essential to the *mitochondria* in our cells for producing *Adenosine Triphosphate (ATP)*. The *Ubiquinol* version of *CoQ10* is highly *bioavailable* and easily crosses the blood-brain barrier. As a nootropic, *CoQ10* not only helps produce *ATP* in brain cells, but protects against free radical damage as well.[ii]

CoQ10 improves athletic performance, works as an antioxidant, and battles *fatigue* and *depression*.

A daily dose of *CoQ10* of 100 mg from a highly *bioavailable* form like *Ubiquinol* is best.

Learn more about CoQ10 & Ubiquinol

Creatine

Creatine is made in the liver, and acts as fuel cells for your cells. Including brain cells. It provides energy on demand. You could say *creatine* is the ultimate nootropic.

Creatine is a popular supplement for athletes because it *boosts physical performance* by going directly to the muscles needing fuel. Vegetarians and vegans typically have lower levels of *creatine* in the body than meat eaters.

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Creatine is also favored by nootropic users because of the fuel demanded by brain cells when using racetams. After crossing the blood-brain barrier, creatine binds to *phosphate*. *Creatine phosphate* in turn fuels energy consumption by the brain.

So *creatine* is good for *muscle fuel*, and *mental performance*. It helps boost *memory*, reducing brain *fatigue*, improve *mood*, is *anti-aging* and a *neuroprotectant*.

Dosage recommendation vary widely based on personal preference and physical demand. It goes from 200 mg to 25 grams daily. As a nootropic the most common *creatine* dosage is 3 – 5 grams per day.

Learn more about Creatine

DHA (Omega 3)

Your brain is 60% fat. So it would be safe to say that to maintain and excel mentally, our brain needs a good supply of healthy fats.

The two most studied *omega-3* fatty acids are *eicosapentaenoic acid (EPA)* and ***docosahexaenoic acid (DHA)***.

DHA makes up a large portion of brain *gray matter*. Brain fat forms cell membranes. And plays a vital role in how our cells function.

Neurons are also rich in *omega-3 fatty acids*. *DHA* is a main component of brain *synapses*. (A synapse is the part of a brain cell that helps a *neuron* pass an electrical signal to another neuron).

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Low levels of *omega-3 fatty acids* result in *ADD, anxiety, depression, obesity, suicide*, and an increased risk of Alzheimer's disease and dementia.

Diets rich in omega-3's help balance emotions and boost mood because *DHA* is a main component of the brain's synapses.

Eating fish helps cognitive performance because fish, krill, and other marine life are high in *Omega-3's*. Other foods high in *Omega-3* fatty acids include anchovies, broccoli, Brussel sprouts, cabbage, cauliflower, flaxseed, soybeans, walnuts, spinach, and tofu.

Ideal daily dosage for Omega-3's should include a least 1,000 mg of *DHA*.

Learn more about DHA

DHEA

DHEA (Dehydroepiandrosterone) is a hormone produced by your adrenal glands, brain, and testes in men. *DHEA* levels decline with age right along with *testosterone*. The result is deceased *motivation, cognition, depression, fatigue*, and loss of *libido*.

DHEA as a nootropic has a noticeable effect on increasing *motivation*, and a general sense of well-being.

Remember that *DHEA is a steroid hormone* and does increase

estrogen. In men, prolonged elevated estrogen levels have negative effects like abdominal fat, and male breasts.

Dosing *DHEA* is 25 mg per day, and you may want to cycle one month on and one month off. But please get your *estradiol* & *DHEA* levels checked if you're going to supplement with *DHEA*.

Learn more about DHEA

DMAE

DMAE (Dimethylaminoethanol) naturally occurs in your brain. *DMAE is not* a precursor to acetylcholine as reported on some other nootropic sites. *DMAE prevents choline metabolism by cells which results in a boost in choline levels in the brain*.

DMAE as a nootropic helps increase *alertness*, boosts *mood* and *memory*. Studies show that *DMAE* supplementation may help improve mood.[iii]

Use of *DMAE* as a nootropic has mixed reviews in the neurohacking community. Some report benefits to increased energy. And others say it causes *hyperactivity*, loss of focus and motivation, and depression. Most of the negative effects of *DMAE* come from higher and prolonged dosing.

Like all nootropics, and particularly with *DMAE*, experiment to find out *if* it's right for you. Start out with a smaller 50 mg dose and go as high as 200 mg per day while looking for your sweet spot.

And check out the "Dosage Notes" and "Side Effects" in the extended article (click the link directly below) for more on *DMAE*.

Learn more about DMAE

5-HTP

5-HTP (5-Hydroxytryptophan) is a serotonin *precursor* in the brain. It is a naturally occurring byproduct of the amino acid tryptophan. Serotonin is a neurotransmitter that helps communication between neurons.

5-HTP as a nootropic easily crosses the blood-brain barrier. And once it converts to *serotonin* can help improve *mood*, control *behavior* and *appetite*, and help you *sleep*.

Research shows *5-HTP* can help in *impulse control* and *balance out moods resulting in less anxiety*.[iv] And even *reduce panic attacks*.

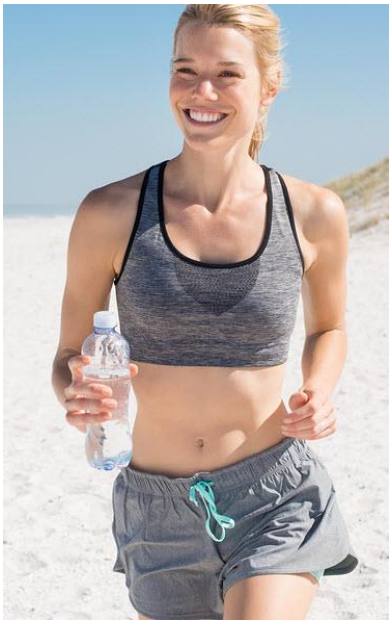
Dosing of *5-HTP* is 50 mg 3-times per day for *up to 2 weeks*.

See the extended article for Nootropics Expert recommendations and *warnings* on supplementing with *5-HTP*.

Learn more about 5-HTP

Forskolin (Coleus root)

Forskolin is a plant native to south Asia. And has a limited nootropic track record. It first became popular when included in the open source stack *CILTEP*.



Forskolin combined with artichoke extract boosts cAMP levels in the brain. Leading to improved *memory*, and *ability to reason*.

Common dosage of *Forskolin* is 150 – 250 mg per day.

Learn more about Forskolin

GABA

GABA (gamma aminobutyric acid) is an *amino acid* and neurotransmitter produced by *glutamate* in your brain. *GABA* in the brain is considered an *inhibitory* neurotransmitter which means is prevents other *neurotransmitters* from being released. Resulting in an *anti-anxiety* and *calming* effect.

There is considerable debate on whether ***GABA*** as a supplement can *cross the blood-brain barrier*. But many find it effective for relieving *anxiety, stress* and for *sleep*.

Adding a *phenyl group* to *GABA* (called Phenibut (β-Phenyl-GABA)), this derivative *is able* to enter your brain and lower levels of *anxiety* and *stress*.

Typical recommended dosage of *GABA* is 250 - 500 mg per day.

Learn more about GABA

Ginkgo Biloba

Ginkgo Biloba comes from the leaves of the *Ginkgo* tree native to China. The leaves have been used for thousands of years to boost mental *alertness*, improve *cerebral circulation*, and overall brain function.

As a nootropic, ***Ginkgo*** has been shown to be particularly effective for elderly memory loss, for slow thinking and reasoning, and tinnitus. One study shows *significant improvement in Parkinson's and Alzheimer's patients*.[v]

Most noticeable in adults suffering cognitive decline, *Ginkgo Biloba* seems to improve *short-term memory* and *recall*. It's also effective in reducing *stress and anxiety* and for *boosting mood*.

Dosage of *Ginkgo Biloba* is typically 120 – 240 mg per day with food. Caution should be taken because *Ginkgo* can increase *internal bleeding* in some users (see "*Side Effects"* in the extended article on NootropicsExpert.com by clicking the link below).

Learn more about Ginkgo Biloba

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Ginseng

Ginseng can improve symptoms of *anxiety*, and boost *attention*, *concentration*, and *memory*. Nootropic users above age 40 find the most benefit in *Ginseng*.

Panax ginseng is native to southeast Asia. Other species include ***American Ginseng*** and *Siberian Ginseng*.

Each have unique characteristics. *Panax Ginseng* is preferred as a nootropic, and is used as a *memory* booster, improves *mood*, lower *anxiety* levels, and boosts *stamina* and *endurance*.

Look for *ginseng* extract of at least 3-5% *ginsenosides*. A good starting dose is 100 mg per day.

Learn more about Ginseng

Gotu Kola

Gotu Kola (Centella asiatica) is one of the most important herbs in ancient Ayurveda medicine. *Ayurveda* uses it to reduce *anxiety*, reduce *fever* and treat *skin conditions*. It improves *circulation* and promotes *longevity*.

Unfortunately, our brains also shrink as we age. Brain growth factor seems to diminish. But clinical trials show *Gotu Kola extract* can help *spur growth in brain cells*.[vi]

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Users report *gotu kola* is at least as effective – perhaps even more so – in reducing anxiety and relieving stress than *Ashwagandha* and *Phenibut*.

As an extract, *Gotu Kola* dosage is 10 drops or 10-20 ml per day. As a dried herb make a tea of the dried leaf and use 3-times daily.

As a powdered herb take 400-600 mg of *Gotu Kola*, three times per day.

Learn more about Gotu Kola

Holy Basil (Tulsi)

Holy Basil (Tulsi) or "*The Incomparable One*" is the most sacred plant in the Hindu religion. And has been valued for millennia by Ayurveda, Greek, Roman, Siddha and Unani because of its benefits for mind, body and spirit.

Holy Basil is a *monoamine oxidase inhibitor (MAOI)* which increases levels of *dopamine*. And it inhibits *acetylcholinesterase (AChE)* which boosts *acetylcholine*.

Holy Basil reduces spikes in *cortisol* induced by *chronic stress*. And is particularly effective in *reducing stress caused by loud noise*.

Holy Basil is also a potent *antioxidant*. It's a natural *COX-2* inhibitor. And the compounds including *eugenol, cirsilineol, cirsimaritin, isothymusin, isothymonin, apigenin*, and *rosmarinic acid* in *Holy Basil* have been compared to the *pain-relieving* activity of ibuprofen, naproxen, and aspirin.

Many neurohackers use *Holy Basil* as an *anxiolytic* and find it *as effective for calming anxiety as some prescription anti-anxiety meds*.

Recommended dosage of *Holy Basil extract* is 300 – 2,000 mg per day.

Learn more about Holy Basil (Tulsi)

Huperzine-A

Huperzine-A is a natural compound extracted from the Chinese club moss *huperzia serrata*. *Huperzine-A* is an *acetylcholinesterase* inhibitor which means it boosts levels of the neurotransmitter acetylcholine in your brain.

Huperzine-A helps promote *memory* by increasing *acetylcholine* levels. Users report improved *memory, retention, cognition*, and *lucid dreaming*.

Dosage of *Huperzine-A* is 200 mcg every 2 days or twice per week.

Learn more about Huperzine-A

Iodine

Iodine *is a cognitive enhancer* not talked about much in the nootropic community. *Iodine* deficient populations are up to 13.5 IQ points less than normal populations.[vii]

Since *iodine* has largely been removed from table salt in some countries, many suggest an *iodine deficiency epidemic*. Particularly in countries like the United States.

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Iodine is essential to a healthy thyroid. Remember, your thyroid is right next to your brain. *Iodine deficiency* during early childhood results in profound intellectual disability.

The thyroid hormones T_4 and T_3 are synthesized from iodine and tyrosine. Which regulate processes like growth and metabolism. Thyroid hormones target organs like the brain where they regulate *gene expression*. And help protect brain cells from toxins.

Ask anyone suffering from *hypothyroidism* about *brain fog*!

Iodine dosage for a healthy adult is up to 1 mg per day. Supplemented with *selenium*. Brazil nuts are a great source of *selenium*. Two Brazil nuts can give you your daily dose of *selenium*.

Learn more about Iodine

Kava Kava

Kava (piper methysticum) is native to the South Pacific. Pacific Islanders traditionally use the plant for its *sedative* effects.

Kava can help to reduce *anxiety*, improve *mood*, and promote an overall sense of *well-being*. And unlike benzodiazepines, *kava* does not impair cognitive function. In fact, studies show kava may boost cognitive function.[viii]

Of the 105 varieties of kava, *Noble* and *Tudei Kava* has been traditionally used in the South Pacific Islands. And have been safely consumed for hundreds of years.

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Daily dosage of *Kava Kava* is 250 – 500 mg. Some experts suggest taking less as it will have less impact on your liver. Experiment to find out what works for you.

Learn more about Kava

Kratom

Kratom *(Mitragyna speciosa)* is the leaf of a tropical deciduous tree within the coffee family (Rubiaceae) native to Southeast Asia.

Mitragynine and *7-hydroxymitragynine (7-HMG)*, unique to *Mitragyna speciosa*, are the two alkaloids mainly responsible for the effects of *Kratom*. And are agonists of the *μ-subtype opioid receptor (MOR)*. Which provides its powerful pain-relieving action.

5-HT_{2a} (subtype of serotonin receptor) and postsynaptic α_2–*adrenergic* receptors, as well as neuronal Ca^{2+} *channels* are also involved in the unique mechanism of action of *mitragynine*.

Kratom's effect on *serotonin* and *norepinephrine* help improve *mood* and lower *anxiety*. At lower doses, *Kratom* is used for its *stimulant* effects and to combat *fatigue*.

At higher dosages, *Kratom* can have more of a *sedative* and *anxiolytic* effect. And has also been used in traditional and natural medicine as *an alternative to opiate medications*.

Learn more about Kratom

Lion's Mane

Lion's Mane Mushroom is an ancient Chinese remedy for improving cognitive performance, and overall health.

Unlike other nootropics which often modulate neurotransmitters, *Lion's Mane* goes to the root of the problem.

It prevents and treats nerve damage by boosting *Brain Nerve Growth Factor*, and *neurogenesis*.

Lion's Mane can help improve *focus* and *attention*, boost *thinking*, *repair brain cells*, help *depression* and *anxiety*, and manage other neurological problems like Alzheimer's, dementia, Parkinson's, and muscular dystrophy.

Lion's Mane Mushroom dosage largely depends on the strength of the extract. And amount of *polysaccharide* content. Try 500 mg to 3 gm per day depending on extract concentration.

Learn more about Lion's Mane

L- Carnosine

L- Carnosine is an amino acid or dipeptide found in your kidneys, liver, muscles and brain. The two amino acids are *beta-alanine* and *L-histidine*.

L-Carnosine works in the body as an antioxidant scavenging for *free radicals*. The concentration in muscles correlates with life span so it's considered a powerful *anti-aging* supplement.

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L-Carnosine is involved in neurotransmission, binds to and helps remove *AGEs (Advance Glycosylation End-products)*, helps in the treatment of Alzheimer's, and assists in removing *heavy metal toxins*.

While you may not see profound and immediate nootropic benefits using *L-Carnosine*, hundreds of studies prove its benefits for long-term health and anti-aging.

Recommended dosage of *L-Carnosine* is 1,000 mg per day. Bodybuilders and athletes prefer supplementing with *beta-alanine* – up to 3.2 grams per day. (Do not confuse ***L-Carnosine*** with *L-Carnitine*).

Learn more about L-Carnosine

L-Dopa (Mucuna Pruriens)

L-Dopa (L-3,4-dihydroxyphenylalanine) is a precursor to the formation of dopamine, norepinephrine, and epinephrine in your brain. In your body, ***L-Dopa*** is synthesized from the amino acid L-tyrosine.

Dopamine deficiency is directly correlated with Parkinson's Disease.

L-Dopa easily crosses the blood-brain barrier when used as a nootropic supplement. *L-Dopa* can increase *libido*, *testosterone*, enhances

memory and learning *retention*. It is also reported to increase *human growth hormone (HGH)* levels.[ix]

You can find extracts of *Mucuna Pruriens* up to 98% pure *L-Dopa* taken from velvet bean or cowitch. Dosage depends on the strength of the extract and can range from 150 mg to 1 gm per day.

Learn more about L-DOPA (Mucuna Pruriens)

Lemon Balm

Lemon Balm (Melissa officinalis) is used for its anti-anxiety effects. This plant from the mint family has a lemony scent and is native to the Mediterranean region.

Rosmarinic acid, a compound found in *lemon balm*, inhibits the *GABA transaminase* enzyme. Which in turn helps maintain adequate levels of *GABA* in your brain. Resulting in a calming effect.

As a nootropic, *lemon balm* is most commonly used for *stress* relief, and reduction of *panic attacks*. *Lemon balm* also helps with *focus, concentration, reduces irritability and depression*, and improves *memory and learning retention*.

Dosage of *lemon balm* is 1 – 2 grams of dried lemon balm leaf as a tea, or 300 – 600 mg of lemon balm *extract* as a nootropic supplement.

Learn more about Lemon Balm

L-Glutamine

As a nootropic, ***L-glutamine*** increases the effectiveness and activity of neurotransmitters. Helping with improve *focus, energy, cognition* and *retention* when *learning*.

L-glutamine in your brain works by boosting *glutamic acid* and *gamma-aminobutyric acid (GABA)*. It also helps detox toxins and *free radicals* from your brain.

And by increasing *GABA* levels, *L-glutamine* can lead to *reduced stress and better sleep*.

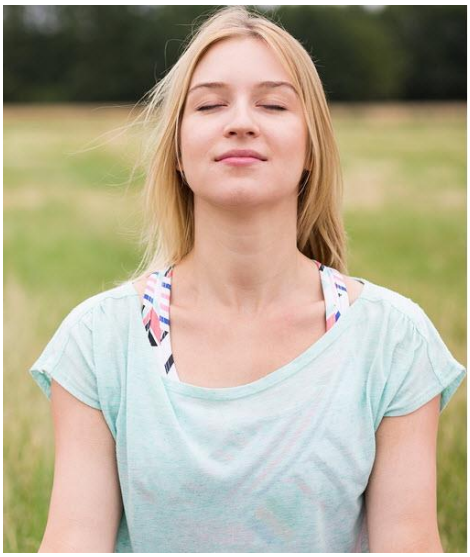
Dosage of *L-glutamine* varies from 500 mg all the way to 10 or 20 grams. For nootropic use it's typically on the lower end of the scale. Experiment with smaller amounts and move up as needed to see what works best for you.

Learn more about L-Glutamine

Lithium Orotate

Lithium retains a grim and undeserved reputation. Because it has been associated with treating serious mental issues like *bipolar disorder* and *mania*.

It turns out that the psychotic medication *lithium* isn't even a drug. It's actually an essential *trace mineral* required to maintain optimal cognitive function.



Studies from around the world have demonstrated the critical health benefits of lithium.

One study using data from 27 Texas counties from 1978 – 1987 found that *rates of suicide, homicide and rape were significantly higher in counties whose drinking water contained little or no lithium*.

As a nootropic, micro-dosing with *Lithium Orotate* comes with a host of benefits. Lithium calms *mania* and *mood swings* by decreasing the sensitivity of norepinephrine receptors. It upregulates *brain-derived neurotrophic factor (BDNF), nerve growth factor (NGF)*, *neurotrophin-3 (NT3)* and their receptors.

And *lithium* stimulates the proliferation of *stem cells* in the brain. All boosting neurogenesis and assisting in repair from all types of brain injury.

Lithium also increases brain *gray matter*, increases *DNA* replication for *neurogenesis*, prevents *apoptosis*, increases *N-acetyl-aspartate (NAA)*, inhibits *beta-amyloid* secretion and protects against damage once it's formed, chelates aluminum, and protects against glutamate toxicity.

Lithium Orotate recommended dose is 5 mg two or three times per day. Or *as needed*.

Learn more about Lithium Orotate

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L-Theanine

L-Theanine, which naturally occurs in green tea and oolong tea, is an amino acid. *L-Theanine* is used as a nootropic for *anxiety, learning, mood*, and *focus*.

Since it easily crosses the blood-brain barrier, *L-Theanine* works quickly to increase dopamine and serotonin. The "feel good" neurotransmitters in your brain.

Experienced nootropic users often stack *L-Theanine* with *caffeine* for a synergistic effect to promote cognition, *motivation* and *attention*. As a bonus, the *caffeine jitters* are reduced when combined with *L-Theanine*.

Dosing *L-Theanine* as a nootropic is 250 – 500 mg per day and it is water soluble.

Learn more about L-Theanine

Maca

Maca is a native Peruvian plant growing over 4,000 m (13,000 ft) high in the Andes mountains.

Maca contains significant amounts of the amino acids arginine, serine, histidine, aspartic acid, glutamic acid, glycine, valine, phenylalanine, tyrosine, and threonine.

Natives in the central Peruvian Andes traditionally had their children eat *Maca* to *improve their performance in school*. Likely due to Maca's ability to boost *acetylcholine* and act as an *antioxidant*.

Maca is used as a sports supplement by strength and endurance athletes to *improve trial performance*. Likely due to better energy *metabolism* and improved *antioxidant* status.

As a nootropic, *Maca* is known for elevating *mood*, increasing *energy* and *endurance*, boosting *learning and memory*, increasing *libido*, and is a potent *neuroprotectant*.

The purest *Maca* comes from high in the Andes of Peru. But *Maca* is also popular now in China and some *Maca* for export comes from that country. And *should be avoided*.

Recommended dosage of *Maca* for cognitive benefit is up to 5 grams per day.

Learn more about Maca

Magnesium

Magnesium doesn't get the respect it should as a nootropic. The lack of adequate levels of magnesium in your body can result in *brain fog*, *anxiety,* and *depression*.

Magnesium deficiency is nearly epidemic in our society because it is sorely lacking in our food supply.[x] [xi]

Plasticity of neuron synapses is affected by the presence of adequate *magnesium* in brain cells. It helps improve memory and cognition. And reduces *anxiety*, *brain fog* and *depression*.

Magnesium can also help maintain healthy blood pressure and help prevent sudden heart attack and stroke.

Most forms of Magnesium don't work well as a nootropic supplement because they can't easily cross the blood-brain barrier.

A new form of magnesium called ***Magnesium-L-threonate*** is recommended for nootropic use. And is a great addition to any nootropic stack.

A typical adult dosage of *magnesium-L-threonate* is up to 1 gram per day.

Learn more about Magnesium

Medicinal Mushrooms

Mushrooms have been used by humans for thousands of years as food and medicine. Of 2,000 edible mushrooms, *270 species* have been identified and tested as therapeutic and preventive agents that ensure and boost human health.

But of those 270 species, *only 6 are considered nootropics*. And provide some benefit specifically for your brain.

And those mushrooms include: *Chaga, Cordyceps, Lion's Mane, Maitake, Oyster* and *Reishi*.

Each of these nootropic mushrooms offer a distinct benefit to your nootropic stack. Clinical studies show they increase *neurotransmitter* levels, repair brain cells, improve cognitive performance and longevity, and reduce *anxiety* and *depression*.

But the *medicinal mushroom* market is fraught with peril. And many manufacturers pass on *inferior product to unsuspecting and uneducated customers*.

See my full review for the benefits that each of these *nootropic mushrooms* offer. And learn how to distinguish between *genuine mushrooms* and *powdered extracts that offer zero benefit*.

Once you learn how to read the label, you'll have more confidence knowing you're buying the best *nootropic mushrooms*.

Learn more about Medicinal Mushrooms

Methylene Blue

Methylene Blue as a nootropic will likely *feel* different than any other nootropic on this list. It's the first *synthetic* drug ever developed. First used to treat malaria back in 1891.

Methylene Blue assists *brain cell respiration* by increasing intracellular oxygen. It donates electrons to the *electron transport chain* within mitochondria which helps create the ATP.

Methylene Blue is a potent *antioxidant*. It binds to *superoxide* created during metabolism within brain cells. Reducing it to water and stopping the oxidative cascade at its very beginning. *Before it gets a chance to do damage*.

And Methylene Blue inhibits *monoamine oxidase (MAOI)* and *acetylcholinesterase* activity which increases levels of *catecholamines* and *acetylcholine*.

Boosting serotonin and norepinephrine levels. Affecting anxiety, depression, energy and memory.

Recommended safe dosage based on clinical studies ranges from 0.5 – 4 mg/kg of body weight. It's *water-soluble* so you don't have to take it with a meal or healthy fat.

But please see the "Side Effects" section in the main Methylene Blue review. Because you don't want to take it with antidepressant or anti-anxiety medications. Or you put yourself at risk of *Serotonin Syndrome*.

Learn more about Methylene Blue

Melatonin

A good night's sleep is about the nearest we can get to the perfect nootropic. ***Melatonin (N-Acetyl-5-Methoxytryptamine)*** is a hormone made in your *pineal gland*. And it's this hormone your body produces when it's time to go to sleep.

Melatonin levels are low during the day. And peak at about 2 – 3 AM depending on your age. After that it steadily declines until morning.

When the timing is off and *melatonin* drops below optimal levels, you know you're getting old(er). Or your *pineal gland* is not working optimally.

When the *melatonin* cycle is disrupted by jet-lag, aging or stress – your ability to think clearly, memory, and decision-making abilities can suffer.

Don't take *melatonin* during the day or it will disrupt your natural circadian rhythm. *Melatonin* as a nootropic supplement taken 1 1/2 hours before bed is ideal.

The appropriate dose of *melatonin* can vary widely from person to person. Start with 1 – 3 mg, 90 minutes before bedtime. See how you feel. And if you readily fall and stay asleep until morning.

Then adjust your dose from there – up or down. Most people don't need any more than 3 – 5 mg of *melatonin* per night. Some even less.

Learn more about Melatonin

N-Acetyl L-Cysteine

N-Acetyl L-Cysteine (NAC) is an amino acid that regulates the amount of glutamate and dopamine in your brain. *NAC* is the precursor to glutathione which reduces *free radicals* in your brain.

N-Acetyl L-Cysteine is used as an anti-dote to Tylenol (acetaminophen) overdose and carbon monoxide poisoning.

NAC is also used to treat ADHD, Bipolar Disorder, prevent alcoholic liver damage, Alzheimer's Disease, eliminate heavy metals, and depression.

N-Acetyl L-Cysteine (NAC) dosage as a nootropic is 600 mg up to three times per day which helps *focus, motivation* and *concentration*.

Learn more about N-Acetyl L-Cysteine

N-Acetyl L-Tyrosine

N-Acetyl L-Tyrosine (NALT) is a highly bio-available form of the amino acid *L-Tyrosine*. Your brain uses *L-Tyrosine* to produce *dopamine*. And the neurotransmitter *norepinephrine* which affects alertness, memory, and mood.

Dopamine is involved with *libido, memory, focus, goal-oriented concentration*, is a *mood elevator* and *anti-depressant*.

Norepinephrine helps with *alertness, working memory, focus*, and *executive function*.

L-Tyrosine supports healthy *glandular* function and stress response because it helps with the synthesis of thyroid hormone and epinephrine (adrenalin).

Studies have found *tyrosine* to be useful for cold, fatigue, prolonged work, stress, sleep deprivation, and those suffering with hypothyroidism. Particularly studies within the military.[xii]

The typical dose for *N-Acetyl L-Tyrosine (NALT)* is 300 – 500 mg dosed 2 or 3-times per day. Experiment to see how you respond during your day.

Learn more about N-Acetyl L-Tyrosine

NADH

NADH (Nicotinamide Adenine Dinucleotide + Hydrogen) boosts alertness, improves mental energy and elevates mood. NADH is an antioxidant coenzyme related to the *B-Vitamin* family.

This *coenzyme* is used in the formation of *ATP*, the energy source produced within the mitochondria of your cells. It's most highly concentrated in your heart and brain. Both power-hungry organs.

NADH seems to decline with age and is associated with cell damage and accelerated cellular aging.

NADH also helps boost the production of dopamine and norepinephrine. This allows for greater mental performance including better memory, alertness, and energy.

Dosing of *NADH* is 5 – 20 mg per day. *NADH* for *nootropic benefit* seems to be 10 mg per day.

Learn more about NADH

Nefiracetam

Nefiracetam is one of the newer nootropics in the *racetam family* of cognitive enhancers. Developed in Japan, *Nefiracetam* is used to treat *learning and memory* deficit diseases.[xiii]

Nefiracetam is structurally similar to *Aniracetam* but considered much more potent. This *fat-soluble* nootropic easily crosses the blood-brain barrier. It works in the hippocampus area of your brain to boost *memory* and *recall*.

Nefiracetam is used clinically to treat and *prevent seizures*, *severe depression*, and has shown *neuroprotective* qualities.

Dosing *Nefiracetam* is based on body weight. Acceptable dosage levels according to limited reported user experience is 100 – 900 mg per day. Most nootropic users experience some benefit in the lower range.

Learn more about Nefiracetam

Nicotine

Nicotine molecules attach to *nicotinic acetylcholine receptors* in the brain. And seem to work by regulating other receptor systems. One obvious clue to how this works – nicotine tends to make you more alert. And has a calming effect.

The primary neurotransmitter that *nicotine* boosts is dopamine. Which may be the reason why it's so addictive. As a result, it can also help prevent some diseases.

Dozens of clinical studies have been published showing *nicotine* effective in treating *Parkinson's, ADHD, Tourette's, schizophrenia*, and other neurological disorders.

Jennifer Rusted, professor of experimental psychology at Sussex University in Britain says, "*nicotine* is the most reliable cognitive enhancer that we currently have".[xiv]

Turns out the addictive quality of using tobacco isn't entirely caused by *nicotine*. But by the interplay with all the other compounds found in a cigarette.

The key to using *nicotine* as a nootropic is NOT by smoking a cigarette. Use a *nicotine patch* or *lozenge* instead.

Recommended dosage of *nicotine* as a nootropic is 1 – 2 mg, preferably used sublingually.

Learn more about Nicotine

Noopept

Noopept is a peptide-derived nootropic related to the racetam family. Much more potent than *Piracetam* (up to 1000X), its mechanism of action in your brain is similar to other racetams.

Noopept seems to stimulate *dopamine, nicotinic*, and *serotonin* receptors. It boosts *cognition, memory, logical thinking*, improves *reflexes*, and improves *mood*.

Noopept has also been shown to increase brain *Nerve Growth Factor*.

Dosage recommendations for *Noopept* are 10 – 30 mg per day. And should be stacked with a good choline source like Alpha GPC or CDP Choline.

Learn more about Noopept

Oat Straw

Oat Straw (avena sativa) comes from green oat grass. You may have heard the term "*sowing your wild oats*". *Oat Straw* increases *luteinizing hormone* in your body and is the basis for that saying. *Luteinizing hormone stimulates testosterone production*.

Oat Straw is a MAO-B inhibitor and increases *dopamine* levels in the brain. As a nootropic, *Oat Straw* can improve *attention, cognition,*

concentration and focus. All benefits experienced when your *dopamine* and *testosterone* levels are optimized.

Oat Straw extract dosage is 800 – 1600 mg per day.

Learn more about Oat Straw

Oxiracetam

Oxiracetam is from the racetam family of nootropics. And up to 5-times more potent than *Piracetam*. *Oxiracetam* stimulates *acetylcholine* use in the brain by working with the *AMPA* and *NMDA receptors.*

Oxiracetam can boost *focus, memory, mental energy, recall*, and improves *fluidity of thinking*. It's most often stacked with a choline supplement because it boosts the use of *acetylcholine* in your brain.

Oxiracetam can be paired with other nootropics in a stack. Recommended dosage is 750 – 1,500 mg per day, and preferably dosed twice per day.

You should use *Oxiracetam* with a good choline source like Alpha GPC or CDP Choline to avoid fatigue or headache when using it.

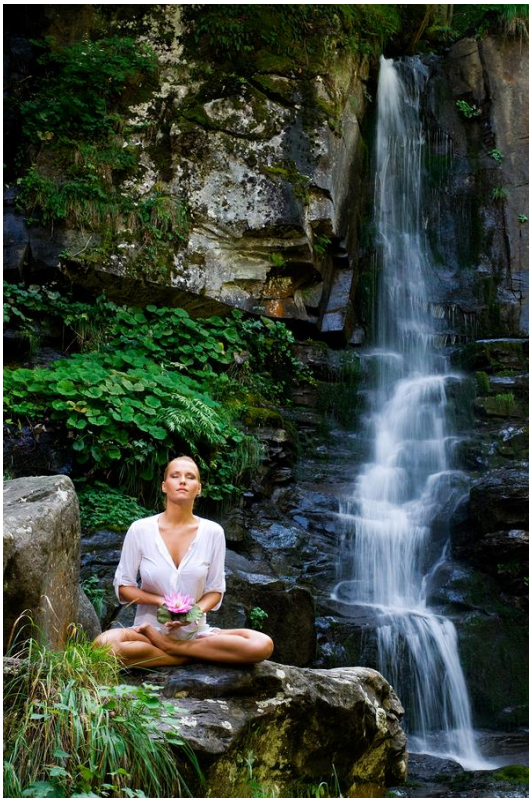
Learn more about Oxiracetam

Phenibut

Phenibut was developed in the Soviet Union in the 1960's. *Phenibut* is a GABA agonist and primarily binds to the $GABA_b$

receptor. *GABAa* receptor agonists include alcohol and benzodiazepines.

Phenibut can have a sedative effect and has strong *anti-anxiety* qualities. It can be used to combat depression, improve mood, cognitive function and motivation.



A safe starting dose of *Phenibut* is 200 – 300 mg. Take it on an empty stomach and expect to feel its full effect in 2 – 6 hours.

Experienced *Phenibut* users will dose as high as 1.5 grams per day (NOT recommended when you're starting out).

Learn more about Phenibut

Phenylethylamine (PEA)

Phenylethylamine (PEA) is a trace amino acid synthesized from Phenylalanine in your brain.

PEA is an endogenous (natural) *amphetamine* in your brain. And its structure is the basis for dozens of amphetamine-based drugs including Adderall (amphetamine), appetite suppressants, decongestants, antidepressants, and psychedelics like MDMA and mescaline.

PEA activates *$TAAR_1$* and *$TAAR_2$* receptors which in turn prevent the uptake and *boosts the release of the 'feel-good' neurotransmitters serotonin and dopamine*.

PEA has a very short half-life including as a nootropic because it's quickly degraded by *monoamine oxidase-B*. So many neurohackers stack *PEA* with a *MAO-B inhibitor* like low-dose selegiline (L-deprenyl), hordenine, or Oat Straw.

Dosage of *Phenylethylamine (PEA*) is 500 mg up to 3-times per day*.* More is NOT better of this nootropic and too much can cause severe side effects including rapid heart rate.

Learn more about Phenylethylamine (PEA)

Phenylpiracetam

Phenylpiracetam (Carphedon) is a Russian derivative of *Piracetam*.[xv] And estimated to be 30 – 60 times more potent than *Piracetam*. It was created by adding an extra molecule from the *phenyl* group.

Easily crossing the blood-brain barrier, *Phenylpiracetam* is a fast-acting nootropic. It helps improve *memory, recall, learning capacity, concentration, motivation*, and mental *energy*. And provides a stimulant effect.

Phenylpiracetam also helps reduce *motion sickness*, boosts physical *performance*, is *anti-anxiety*, and improves *tolerance to cold*.

Dose *Phenylpiracetam* 100 mg twice per day. Tolerance is likely so it's best to cycle *Phenylpiracetam*. Use it when needed for a cognitive boost. Or try cycling it one week on and one week off.

Learn more about Phenylpiracetam

Picamilon

Picamilon (nicotinyl-y-aminobutyric acid) is a combination of *niacin (Vitamin B_3)* and *GABA*. It was developed in the Soviet Union as an enhanced form of *GABA* that is capable of crossing the blood-brain barrier.

Once in your system, *Picamilon* separates into *niacin* and *GABA*. And produces a calming and blood vessel dilation effect.

Similar to *Phenibut*, *Picamilon* improves *memory, concentration, motivation, focus*, has strong *anti-anxiety* properties, and can lower blood pressure.

Start with the lowest possible dosage of *Picamilon* and see how you tolerate it. 50 – 300 mg per day is typical. Most nootropic users find somewhere between 50 – 100 mg 2 or 3-times per day is optimal.

It is *water soluble* and should be taken on an empty stomach for quicker action.

Learn more about Picamilon

Pine Bark Extract (Pycnogenol®)

Pine Bark extract is a standardized extract of French maritime pine bark. ***Pycnogenol®*** is a patented water-soluble pine bark extract containing 65-75% *proanthocyanidins*.

Pine Bark extract helps boost *cerebral blood flow* by increasing nitric oxide which helps dilate blood vessels. And helps repair and maintain the health of blood vessel linings.

Pine Bark extract is one of the *most potent antioxidants discovered*. Scavenging *free radicals* to protect *neurons* and other brain cells from *oxidative* damage.

Proanthocyanidins are condensed tannins responsible for the astringent character (pucker factor) of fruits, berries, beans, and tea. *Pine Bark* is higher in these *bioflavonoids* than most other plant sources.

Pine Bark extract is a potent *anti-inflammatory*, and *antioxidant*. And it's not just great for your brain. It's also great for *skin health, DNA, lowering blood sugar levels, improving eye health, helps correct erectile dysfunction, calms PMS symptoms, tames asthma and hay fever, and symptoms of ADHD*.

Dosing *Pine Bark Extract* or *Pycnogenol*® as a nootropic is up to 100 mg 3-times per day.

Learn more about Pine Bark Extract (Pycnogenol®)

Piperine

Piperine is black pepper extract. And is used with other supplements for greater and faster absorption.

But most don't realize that *Piperine* is a nootropic in its own right. This compound has been shown to decrease *anxiety*, improve *mood* and boost *memory*.

Piperine taken with a supplement like curcumin will prevent the liver from breaking down the curcumin before absorption.

Piperine also elevates other supplement levels in your body which can be good in some cases. And detrimental in others.

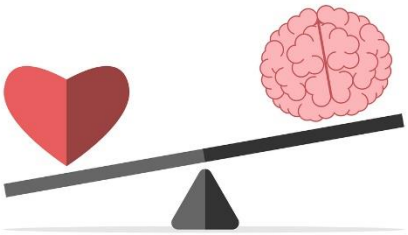
You'll often see ***BioPerine®*** added as a compound in some nootropic or supplement stacks. *BioPerine®* is a patented form of *Piperine* that is said to significantly enhance the *bioavailability* of various nutrients through increased absorption.

Dosage is typically 5 – 20 mg per day of *Piperine*.

Learn more about Piperine

Piracetam

Piracetam was developed by Romanian chemist, and the godfather of nootropics, ***Dr. Corneliu E. Giurgea*** in 1964. This is the first *racetam* ever developed.



Piracetam is a *cyclic derivative* of *GABA* but does not affect the *GABA receptors* in your brain.

Instead, *Piracetam* seems to influence the *AMPA* and *NMDA receptors*. This affects *learning and memory* processes in the brain.

Piracetam also affects the neurotransmitter *acetylcholine* by boosting *ACh* receptors into accepting or being more sensitive to *acetylcholine*.

As a nootropic, *Piracetam* boosts *focus, learning*, and *memory*. *Piracetam* also acts as a neuroprotectant.[xvi]

Dosing of *Piracetam* ranges from 1 – 4.8 grams per day. Dosed throughout the day. If it's your first time using *Piracetam* start at the lower end of the range and work your up. If you're stacking with other *racetams* you'll likely use smaller quantities of *Piracetam*.

Learn more about Piracetam

Rhodiola Rosea

AMPK (adenosine monophosphate-activated protein kinase) is an enzyme found inside each of our cells and works as the body's master regulating switch. *AMPK* is reduced as we age. But it is possible to increase *AMPK* using compounds such as ***Rhodiola Rosea***.

When *AMPK* is "switched on" it triggers the use of stored energy from fats, removes fats and sugars from the blood, boosts *mitochondria* production, reduces inflammation, and takes out the cellular "garbage".[xvii]

When *AMPK* is activated in brain cells it prevents diseases such as Alzheimer's and Parkinson's.

Rhodiola Rosea is a plant that grows at high altitudes and in cold regions of the world. It decreases *depression* and stress-related *mood swings*, reduces *fatigue*, stimulates *energy* and *alertness*, and boosts cognition.

Dosage of *Rhodiola Rosea* is from 150 – 200 mg per day.

Learn more about Rhodiola Rosea

Phenylalanine

Phenylalanine is an *amino acid* that easily crosses the blood-brain barrier. It is not produced by the body so must be obtained by food or in supplement form. Examples of natural sources of *Phenylalanine* are meat and milk products.

Phenylalanine is a precursor, or assists in producing the amino acid tyrosine in your brain. *Tyrosine* then helps in the formation and utilization of neurotransmitters *dopamine, norepinephrine, epinephrine*, and *melatonin*.

Using *Phenylalanine* as a nootropic helps fight *chronic pain*, improves *mood* and boosts *energy*. It also helps in relief from *anxiety*, improves focus and boost *motivation*. DL-Phenylalanine is used to help combat ADHD and Parkinson's.

Phenylalanine dosage is 500 mg up to 3-times per day.

Caution should be observed if you're taking antidepressants.

Learn more about Phenylalanine

Phosphatidylcholine (PC)

Phosphatidylcholine (PC) is a phospholipid with the highest concentration found in the brain and liver. Used as a nootropic supplement, *Phosphatidylcholine (PC)* helps build and repair cell membranes.

When *phosphatidylcholine* is used as a nootropic, it separates into *choline* and *sphingomyelin* in your brain. ***Choline*** is needed for the synthesis of the neurotransmitter *acetylcholine*.

And ***sphingomyelin*** helps in the development and protection of *brain nerve cells*.

When blood levels of *choline* are low, *phosphatidylcholine* is cannibalized from cell walls to make *acetylcholine*. This eventually leads to nasty diseases like *Alzheimer's, poor memory and recall, loss of focus and brain fog*, and more.

To boost *choline* in your brain, you can supplement with *phosphatidylcholine*. As a nootropic supplement, dosing ranges from 1,200 mg – 5 grams per day. If you are new to neurohacking, start on the lower range of the scale and dose 2 or 3 times per day.

Learn more about Phosphatidylcholine (PC)

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Phosphatidylserine (PS)

Phosphatidylserine (PS) is vital for the brain because this *phospholipid* is found in all cells, but most highly concentrated in *brain cell wall membranes*.

Phosphatidylserine helps in the storage, release and activity of *neurotransmitters* and their *receptors*.

Phosphatidylserine production declines as we age. And the reason why supplementation of this nootropic is so critical.

Study after study shows supplementation of *phosphatidylserine* can boost *cognition*, *focus*, memory, and *recall*.[xviii]

Phosphatidylserine (PS) is now derived from plant sources, such as *soybean lecithin*, or *sunflower lecithin*. Plant derived *PS* is equally effective and safer than that derived from animal brain sources.

Dosing *Phosphatidylserine (PS)* is anywhere from 100 – 300 mg per day.

Learn more about Phosphatidylserine (PS)

PQQ

PQQ (pyrroloquinoline quinone) is an enzyme *cofactor*, and the only nutrient known to *facilitate the growth of new mitochondria* in your brain cells.

Researchers found *PQQ* supplementation can boost the production and release of *nerve growth factors* in cells that support creation of

new *neurons* in the brain. And promotes *neuroplasticity* that helps neurons develop the connections needed for *learning and* developing *memories*.[xix]

More studies support *PQQ's* ability to increase *mitochondrial density, reduce inflammation* and *oxidative stress*, and improve learning and memory. As an antioxidant, *PQQ* is thousands of times more potent than *Vitamin C*. Without *PQQ*, *mitochondria* wear out and brain cells age faster.

Supplementing with *PQQ* should give you a *brain energy boost*. And provide *anti-aging benefits*.

Dosing *PQQ* as a nootropic is typically 10 – 20 mg per day.

Learn more about PQQ

Pramiracetam

Pramiracetam is a GABA-like derivative of *Piracetam*. It's said to be 5 – 30 times more potent than *Piracetam*.

Pramiracetam is a fat-soluble nootropic and highly *bioavailable*. It is thought to increase *acetylcholine* receptors in the brain. This *racetam* improves *memory* and *verbal recall*.[xx]

Because *Pramiracetam* stimulates choline uptake in the brain it should be stacked with a good *choline* source like Alpha GPC or CDP Choline.

And it's *fat-soluble* so needs to be taken with food or a good fat source like unrefined coconut oil or MCT Oil.

Optimal dosage of *Pramiracetam* as a nootropic is 1,200 mg divided into 2 or 3 doses during the day.

Learn more about Pramiracetam

Pterostilbene

Pterostilbene is found in cranberries, blueberries, and grapes. And is a compound similar to *Resveratrol*.

Pterostilbene is a very potent *antioxidant*, stimulates *Brain-Derived Neurotrophic Factor*, promotes *neuroplasticity*, is *anti-anxiety*, boosts *dopamine*, and helps *cognition*, *learning* and *memory*.

Pterostilbene works by modifying enzymes linked to glucose levels. It helps reduce blood sugar and cholesterol. Which reduces *oxidative stress* and prevents heart attacks and stroke.

Pterostilbene is believed to be stronger than *Resveratrol* and has better bio-availability. Both compounds work in different ways in a human cell so are often stacked together.

Recommended dosage of *Pterostilbene* is 50 mg per day.

Learn more about Pterostilbene

Resveratrol

A *polyphenol antioxidant* found in the skin of grapes, ***Resveratrol's*** purpose is to protect the grape from *microbial attacks*, cold weather and UV radiation.

Studies have shown *Resveratrol* can also boost *learning, memory* and cognitive power in the hippocampus section of your brain. *Resveratrol* can also improve blood flow and boost brain health.[xxi]

Researchers also discovered *Resveratrol* improves cell survival and neurogenesis in the *hippocampus*, boosting *memory* and *learning*.[xxii]

Recommended nootropic dosage of *trans-resveratrol* is 20 – 250 mg per day.

Learn more about Resveratrol

Saffron

Saffron is the dried stigma of the *Crocus sativus* plant native to the Middle East. And is the world's most expensive culinary spice largely due to the way it must be grown and harvested (by hand).

Saffron has been used for thousands of years as an *anxiolytic*, sedative, and antidepressant.

As a nootropic, *Saffron* is now used for *depression, PMS symptoms, post-partum depression, memory, appetite suppression, energy & stamina in athletes*, and preventing *neurodegenerative diseases like Parkinson's and Alzheimer's*.

Recent studies show *Saffron* as effective as some popular prescription antidepressants. And can even help *alleviate the sexual dysfunction caused by these drugs*.

Saffron extracts (*crocin* & *safranal*) inhibit the uptake of *dopamine* and *norepinephrine* in the brain which *helps improve mood*.

Saffron inhibits the deposit of *amyloid-β* which is associated with the progression of Alzheimer's disease. *Saffron* also inhibits the breakdown of *acetylcholine* (acetylcholinesterase) just like the current medication (donepezil) approved to treat Alzheimer's.

Crocin and *safranal* which are unique carotenoids in *Saffron* make it a potent *antioxidant*. Scavenging *free radicals* which reduces inflammation, preventing *apoptosis*, and protecting brain cells and *mitochondria*.

Recommended dose of *Saffron* is 25 – 30 mg twice per day.

Learn more about Saffron

SAM-e

SAM-e (S-Adenosyl Methionine) is a naturally occurring compound in your body. It is the amino acid *methionine* bound to an *ATP* molecule.

SAM-e helps the process of cell division and repair, and the generation of the neurotransmitters *dopamine, serotonin, norepinephrine*, and *melatonin*. Which is why it's considered a potent nootropic.

SAM-e is used to boost *mood*, relieve *depression*, improve *energy* levels, and reduce symptoms of *fibromyalgia*.

The recommended dosage of *SAM-e* as a nootropic is 400 mg per day.

Learn more about SAM-e

St John's wort

St John's wort (hypericum perforatum) is a plant that has been traditionally used for mood disorders and wound healing. Today it's used mostly as a treatment for *anxiety, depression*, and *stress*.

St John's wort easily crosses the blood-brain barrier. And seems to work preventing the re-uptake of serotonin in the brain.

A word of caution, however. *St John's wort* is a strong *CYP3A4* liver enzyme inducer. This enzyme is responsible for metabolizing psychoactive compounds in the liver. And translates to nullifying the effect of any other nootropics you may be taking.

Dosing *St John's wort* is anywhere from 900 to 1,800 mg per day. But *be careful of using it with any other nootropics or medications*.

Learn more about St. John's wort

Sulbutiamine

Sulbutiamine (isobutyryl thiamine disulfide) is synthesized from Vitamin B_1 (thiamine). It was first developed in Japan to treat beriberi – a *Vitamin B_1 deficiency*.

Superior to *thiamine* as a nootropic because *Sulbutiamine* easily crosses the blood-brain barrier.

As a nootropic, *Sulbutiamine* is taken to boost *mood, memory*, and *motivation*.

A typical dose of *Sulbutiamine* is 400 – 600 mg per day. *Tolerance* has been reported by neurohackers, so it would be wise to *cycle* the use of this nootropic. Skip a couple of days every week.

Learn more about Sulbutiamine

Taurine

Taurine is one of the most abundant amino acids in your brain and body. And a featured ingredient in many popular energy drinks.

As a nootropic, *taurine* functions as a *neurotransmitter* and neuromodulator. It activates GABA and glycine receptors which affects your *mood* and *prevents seizures*.

Taurine is a neuroprotectant because it reduces intracellular free calcium, protects from *mitochondrial dysfunction*, modulates *energy metabolism* within cells, modulates *longevity genes*, inhibits *cellular stress* associated with Alzheimer's, acts as a brain cell health "quality control" and *protects against stroke*.

And *taurine* increases *stem cells* and progenitor cells in your brain which increases brain *cell survival* and *neurogenesis*.

Popular energy drinks such as Red Bull and Monster are not the best sources of *taurine* for the serious neurohacker. You can get the same effect without the negatives by stacking *taurine* with a good *caffeine* source and a *B-Complex* supplement.

Recommended *taurine* nootropic dosage is 500 mg – 2 gm per day.

Learn more about Taurine

Tryptophan

Tryptophan is an essential amino acid and precursor to serotonin, melatonin and niacin (Vitamin B_3) in your body.

Serotonin, the "happiness molecule" relies on an adequate supply of *Tryptophan* for synthesis. Low levels of serotonin in the brain are often due to an inadequate supply of *Tryptophan*.

Tryptophan is found in abundance in oats, bananas, dried prunes, milk, tuna, cheese, bread, chicken, turkey, peanuts, and chocolate. But our bodies often have problems converting this Tryptophan to serotonin in our brain.

Adding *L-Tryptophan* to your nootropic stack can help you boost *serotonin* levels. As a nootropic supplement, *L-Tryptophan* is used to treat *anxiety, ADHD, depression, insomnia, memory loss, pain* and *eating disorders*.

Learn more about Tryptophan

Turmeric

Turmeric (Curcuma longa) is the anti-Alzheimer's spice. *Turmeric* is one of the main spices in curries. In parts of India where curries are eaten most often, Alzheimer's disease is extremely rare.

Turmeric is unique in the ability to reduce *inflammation* common to Parkinson's, Alzheimer's, and brain tumors.

Researchers at the Department of Neurosurgery, University of South Florida College of Medicine in Tampa, FL found that the *curcuminoids* in *turmeric* help break apart the plaque that clogs the brains in Alzheimer's patients.[xxiii]

Turmeric has a combination of *curcuminoids*, volatile oils and proteins that make it anti-bacterial, anti-cancer, anti-inflammatory, and anti-septic.

Some neurohackers maintain that *turmeric* or *curcumin* is the best nootropic. You can increase the *bioavailability* of *turmeric* by combining it with Piperine (black pepper extract) and a healthy fat like olive or *coconut oil*.

To witness the nootropic effects on mood and stress, dose *Turmeric* 2.5 – 4 grams per day.

Some concentrated extracts allow you to get the effects of *Turmeric* in smaller doses. *Curcuma Longa (root) extract* with *95% curcuminoids* is dosed at 750 mg 3-times per day.

Learn more about Turmeric

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Tyrosine

L-Tyrosine is an amino acid produced by *Phenylalanine* in your body. *L-Tyrosine* is directly involved in creating dopamine, norepinephrine, and epinephrine in your brain and body.

L-Tyrosine is also a precursor to thyroxin (your body's main thyroid hormone).

You get *tyrosine* from almonds, bananas, dairy products, eggs, lima beans, oats, poultry, and wheat germ. *Tyrosine* hitchhikes on the back of amino acids like *tryptophan* to cross the *blood-brain barrier* to enter your brain.

N-Acetyl-L-Tyrosine (NALT) is a more *bioavailable* form of *tyrosine* when used as a nootropic.

Neurohackers report that *L-Tyrosine* and especially *NALT* improves *motivation* and *focus* as well as amphetamines like Adderall if you're dealing with ADHD.

L-Tyrosine also plays a role in controlling organs responsible for creating and regulating hormones in your body. Including your adrenal glands, pituitary gland and thyroid.

L-Tyrosine can sharpen *memory*, is *anti-anxiety*, boosts *mood*, and protects brain nerve cells from neurotoxins.

Typical dosage of *L-Tyrosine* is 500 mg – 2 grams per day. Start at the lowest dose and work your way up as you evaluate the effects and benefits of *L-Tyrosine* in your body.

Learn more about Tyrosine

Uridine Monophosphate

Uridine Monophosphate (5'-uridylic acid) is a precursor to *Ribonucleic Acid (RNA)*. *RNA* provides instructions to your *DNA* to help create memory by facilitating connections between brain neurons (synapses).[xxiv]

RNA levels decrease as we age. Supplementing with *Uridine* as a nootropic is *one of the ultimate anti-aging tools to improve memory function*.

The *monophosphate* portion of *uridine* is the transporter that helps *uridine* move through your digestive tract unharmed. And delivers *uridine* across the blood-brain barrier.

Uridine affects the synthesis of phosphatides in the brain which are critical to cellular membranes.

When you increase the *synaptic* proteins in your brain, you boost the number of *synapses*. Improving various aspects of cognition.

Uridine comes from eating broccoli, sugarcane, yeast, liver, and tomatoes.

Uridine supplementation as a nootropic uplifts and stabilizes *mood*, is *anti-stress*, helps *Obsessive Compulsive Disorder (OCD)*, is *anti-anxiety* and helps modulate and normalize dopamine release.

Recommended dosage for you just starting out with *uridine monophosphate* as a nootropic is 150 – 250 mg twice per day. Take it with a good *Vitamin B-Complex* and a 1,000 mg of a high quality *DHA* supplement.

You can take *uridine monophosphate* sublingually for a more pronounced effect. Adding 300 mg of Alpha GPC or CDP Choline is also recommended.

Learn more Uridine Monophosphate

Valerian

Valerian (Valeriana officinalis) is native to Europe and Northern Asia and has been used for millennia for headaches, anxiety, heart palpitations, high blood pressure, irritable bowel, menstrual cramps, epilepsy, and childhood behavior problems.

As a nootropic, *Valerian* is used primarily to *lower stress* and promote a *good night's sleep*.

Valerian helps eliminate stress by boosting *GABA*, serotonin, activating *adenosine receptors*, and reducing the stress hormone *cortisol*.

Valerian promotes sleep by boosting *GABA* and activating *adenosine receptors*. Helping you to relax and *get to sleep faster*. You *sleep better* and *wake up feeling refreshed* with no hangover.

Recommended nootropic dosage for *Valerian root extract* is 200 – 600 mg per day.

Learn more about Valerian

Vinpocetine

Vinpocetine enhances *brain blood flow* by dilating blood vessels. And reducing blood viscosity or thickness.

Vinpocetine is derived from the periwinkle plant. It easily crosses the blood-brain barrier. And helps improve *brain blood supply*, boosts *oxygen* and *glucose* use by the brain, maintains healthy levels of neurotransmitters, and promotes better *concentration, focus*, and *memory*.

Dosing *Vinpocetine* as a nootropic is 10 mg up to three times per day.

Learn more about Vinpocetine

Vitamin B_1 (Thiamine)

Vitamin B_1 (Thiamine) is a required cofactor to produce cellular energy (ATP) from the foods you eat. It's also needed for the synthesis of *DNA* and *RNA*.

Vitamin B_1 is found in lentils, whole grains, pork, red meat, yeast, nuts, sunflower seeds, peas, milk, cauliflower, and spinach.

Vitamin B_1 is *water soluble* and is stored in your body for only 14 days. If you don't get enough *B_1* you'll experience *irritability, confusion*, and *memory problems*. A severe deficiency can manifest itself as *beriberi, pain, heart problems*, or even *paralysis*.

Recommended daily dosage of *Vitamin B_1* is only 1.4 mg per day. You'll experience a nootropic benefit with *Vitamin B_1* at higher doses of 50 – 100 mg per day.

Sulbutiamine is a derivative of *Vitamin B_1 (Thiamine)*. It is more *bioavailable* than standard *Thiamine*. And a far better choice for nootropic use.

Learn more about Vitamin B_1 (Thiamine)

Vitamin B_3 (Niacin)

Vitamin B_3 (Niacin or nicotinic acid) is used in the synthesis of *Coenzyme A* which helps to form *Acetyl-CoA*. *Acetyl-CoA* joins with choline to form acetylcholine.

Vitamin B_3 also supports over 200 other chemical reactions in your body including cellular energy production and fatty acid synthesis.

Food sources of *Vitamin B_3* include beef, poultry, fish, peanuts, and lentils.

Vitamin B_3 (Niacin) is the only *B vitamin* that can be synthesized in the liver from the amino acid tryptophan.

Niacin causes blood vessels to dilate or open up which is especially noticeable near the skin. Taking regular *niacin* as a nootropic supplement can cause a tingling sensation or red flushing of the skin.

1,000 mg of "extended-release" niacin taken 3-times per day can improve *memory* and correct some senility problems. Dosing 50 – 1,000 mg at bedtime taken at bedtime may help you sleep better.

Learn more about Vitamin B_3 (Niacin)

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Vitamin B_5 (Pantothenic Acid)

You need ***Vitamin B_5 (Pantothenic Acid)*** to synthesize *coenzyme-A* for the production of the neurotransmitter acetylcholine.

Your body cannot make *Vitamin B_5* on its own, so it needs to come from food or a supplement. *Vitamin B_5* is found in nearly every food type. But much of the *Vitamin B_5* in Western processed food has been removed during processing.

You may not realize the full benefit of a nootropic stack containing Alpha GPC, CDP-Choline, Acetyl-L-Carnitine or any of the racetams (that boost choline in the brain) without adding *Vitamin B_5.*

Vitamin B_5 also helps support fatty acid synthesis and cellular energy production in your body.

Using *Vitamin B_5* as a nootropic can boost *focus, memory, learning*, and reduce *brain fog*.

Recommended dosing of *Vitamin B_5* is 5 mg per day. But neurohackers suggest a higher dosage like a one-to-two ratio with a *choline* supplement. For example, 250 mg of *Vitamin B_5* with 500 mg of *CDP Choline*.

Learn more about Vitamin B_5 (Pantothenic Acid)

Vitamin B_6 (Pyridoxine)

Vitamin B_6 (Pyridoxine) as a nootropic helps your body make serotonin, norepinephrine and melatonin.

Vitamin B_6 helps control *homocysteine* in your blood. *Homocysteine* is the amino acid associated with heart disease. Your body also requires *Vitamin B_6* to absorb *Vitamin B_{12}* and to make *red blood cells* and cells for your *immune system*.

Symptoms of low *Vitamin B_6* are associated with *irritability, depression, nervousness*, difficulty *concentrating* and *memory loss*.

Dietary sources of *Vitamin B_6* include poultry, tuna, salmon, shrimp, dairy products, lentils, beans, spinach, carrots, bananas, brown rice, and sunflower seeds.

Recommended nootropic dosage of *Vitamin B_6* is up to 100 mg per day. Doses exceeding 200 mg can cause neurological disorders and loss of feeling in your legs.

Learn more about Vitamin B_6 (Pyridoxine)

Vitamin B_8 (Inositol)

Vitamin B_8 (Inositol) as a nootropic can boost serotonin levels which results in *feelings of calm*, heightened *mental energy*, and easy *thought flow*.

But is turns out *Vitamin B_8* is not a true "vitamin" because your body can produce small quantities of this compound on its own.

Vitamin B_8 supports messenger signals throughout the body, including all the messenger signals between neurons in your brain.

Vitamin B_8 (Inositol) is also used to control the symptoms of *Obsessive Compulsive Disorder (OCD)* and *panic attacks*.

Best food sources of *Inositol* are citrus fruits, green leafy vegetables, liver, brown rice, and cereals.

As a nootropic, *Vitamin B_8 (Inositol)* dosage is from 500 – 3,000 mg per day.

Learn more about Vitamin B_8 (Inositol)

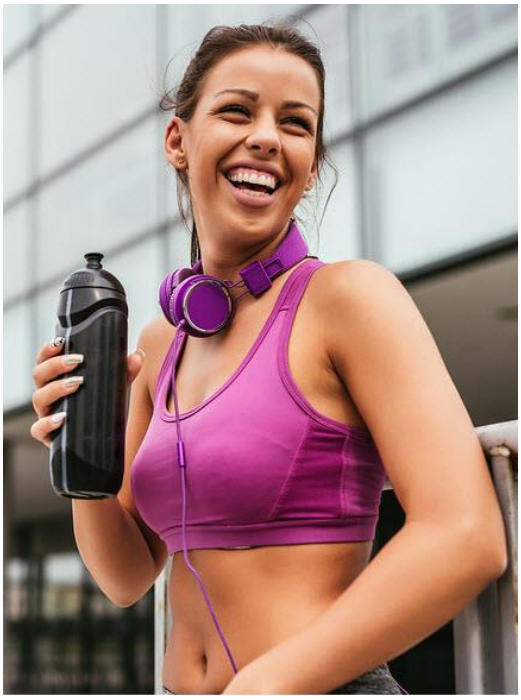
Vitamin B_9 (Folate)

Vitamin B_9 (Folate) as a nootropic helps your brain make dopamine, epinephrine, norepinephrine and serotonin.

As a coenzyme, *Folate* participates in *one-carbon transfers* in the synthesis of *DNA* and *RNA*. And converts *homocysteine* to *methionine* which is used in the synthesis of *S-Adenosyl-methionine (SAMe)*.

Folate is involved in *gene expression*, *amino acid* synthesis, and *myelin* synthesis and repair.

Green leafy vegetables, or '*foliage*' are rich sources of *folate*. And how '*folate*' got its name. You can also get *folate* from citrus fruit juice, legumes, fortified foods (more on this controversy in the extended article), and liver.



Many neurohackers, including doctors and other health professionals confuse ***folate*** with ***folic acid***. They are NOT the same.

In the extended article, we explore the differences between *folate* and *folic acid*. And how *Vitamin B_9 (Folate)* is critical for the fully optimized brain.

As a nootropic, *Vitamin B_9 (Folate)* dosage is 400 mcg per day.

Learn more about Vitamin B_9 (Folate)

Vitamin B_{12} (Cobalamin)

Vitamin B_{12} (Cobalamin) deficiency is common among Western adults.[xxv] *Vitamin B_{12}* plays a key role in the efficient conversion of *carbohydrates* to *glucose* – your cell's source of fuel. It also helps your body to *convert fatty acids* into *energy*.

Brain fog and *poor memory* are two key warning signs your deficient in *Vitamin B_{12}*. Other warning signs include *fatigue, lack of energy, muscle weakness, mood swings, depression, dementia,* and *Alzheimer's*.

Vitamin B_{12} is the largest vitamin we know of, so it's not as easily absorbed as others. As you get older the body loses the ability to transport *Vitamin B_{12}* in the cells in your large intestine where it's pulled into your bloodstream.

The older you get the more difficult it is for you to get ANY B_{12} into your body. So the older you get the more likely you'll need to supplement with *Vitamin* B_{12}.

B_{12} in its *natural form* is only available through *animal food sources* including seafood, beef, chicken, pork, milk, and eggs.

To get an adequate supply of *Vitamin* B_{12} as you get older is really *only available through supplementation*.

Even though B_{12} is *water-soluble*, it doesn't exit your body as quickly as other water-soluble supplements. It's stored in your liver, kidneys, and other body tissues. This means a deficiency *may not show up for a number of years*. But by then it may be too late and *irreversible brain damage* can potentially result.

The best way to dose *Vitamin* B_{12} *(Cobalamin)* as a nootropic is *sublingually* (under the tongue), or a B_{12} shot. The first way is much less expensive and considerably less painful.

Higher quality *Vitamin* B_{12} comes as ***methylcobalamin*** or ***adenosylcobalamin*** which are forms of B_{12} naturally occurring in your body. When looking for a B_{12} supplement or *B-Complex*, choose the one that contains ***methylcobalamin*** for much better absorption and use by your body.

The recommended dosage for *Vitamin* B_{12} deficiency is 2,000 mcg daily of ***methylcobalamin*** for a week, then 1,000 mcg doses of B_{12} once per week for a month. Then your maintenance dose is 1,000 mcg monthly.[xxvi]

Learn more about Vitamin B_{12} (Cobalamin)

Vitamin D

Vitamin D as a nootropic is critical for the synthesis of *GABA*, *glutamate* and *glutamine*, and *dopamine* in your brain.

Vitamin D is involved in *gene expression*, regulation of *neurotrophic factors*, *neuroplasticity*, and brain development early in life. All in *areas of the brain associated with depression*. And the *neuroplasticity* needed for *memory formation* and *retrieval*.

Vitamin D protects your brain from *DNA* damage through prevention of *telomere* shortening and inhibition of *telomerase* activity. And prevents oxidative damage to *DNA*.

Vitamin D_3 is the *fat-soluble* steroid hormone form of *Vitamin D*. The "*sunshine vitamin*" is considered *essential*. Your skin synthesizes *Vitamin* D_3 from *ultraviolet-B (UVB) sunlight*.

As a neurohacker, you should be aware that you are likely deficient in this essential vitamin. *Vitamin D deficiency (hypovitaminosis D)* is an undeclared worldwide pandemic affecting nearly *50% of the population on this planet*.

The preferred method for getting adequate Vitamin D is from UVB sunlight on exposed skin.

But all kinds of things can interfere with getting enough *Vitamin D* from sunlight. Fall and Winter in both hemispheres, cloud cover, smog, skin color, sunscreen, and too much clothing are all factors.

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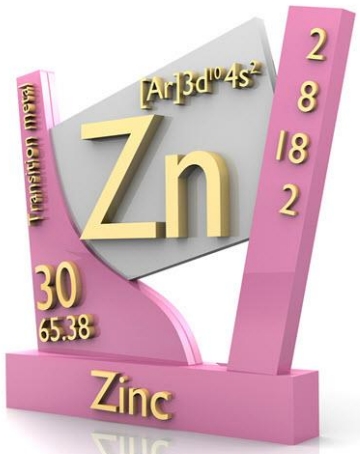
And most of us spend so much time indoors, to get its benefits you should take *Vitamin D_3* as a supplement.

Recommended *Vitamin D_3* dosage is 4,000 IU's per day.

Learn more about Vitamin D

Zinc

Zinc is an *essential* trace element absolutely required for optimal brain health and cognition.



Zinc is required for memory formation. When more *zinc* is available in *neurons*, *neurotransmitters* move more efficiently and stay in play longer.

Zinc helps boost BDNF which is needed for *long-term potentiation* and the formation of *memory*. A deficiency in this mineral results in poor memory.

It is integral to the activity of at least 100 enzymes. And zinc helps lower oxidative stress which protects brain cells and cerebral blood flow. It is part of protein synthesis needed to make *neurotransmitters*. And it's involved in DNA repair, synthesis and cell division.

Zinc deficiency is common even in our Western society because we often don't get enough from food. If you're vegan or vegetarian, you are particularly susceptible.

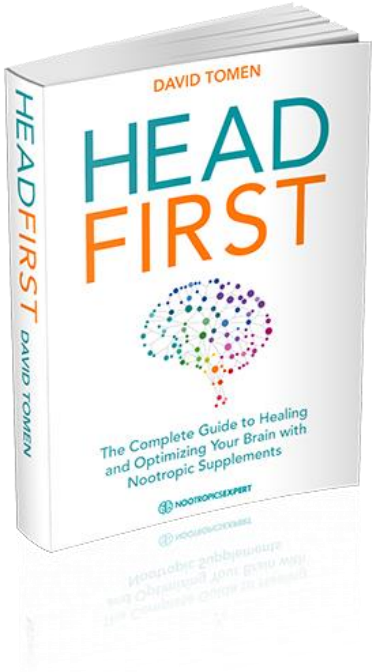
Zinc levels are low in those with *depression*. And *the lower the zinc level, the more severe the depression*.

Recommended *zinc* dosage is 30 mg daily, balanced with 2 mg of *copper*.

Learn more about Zinc

Tap into The Full Power Of Your Brain

In ***Head First***, nootropicsexpert.com author David Tomen provides the latest information on all the most popular nootropic supplements used today. Nearly *600 pages* packed with information in language that is easy to read and understand by both the layman and medical professional.



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Through studying the clinical data and experimenting with dozens of nootropic supplements, the author has successfully restored his own health. With cognition, memory, and mood better than before.

Through his writing and videos, the author is now empowering people world-wide to take control of their own cognitive health.

Fans of David's work and contribution to the neurohacking community are encouraging their doctors and psychiatrists to make ***Head First*** part of their library.

When you take care of your ***Head First***, everything else falls into place. Your full potential is within reach at last. Today is the day to go ***Head First***!

Click here and get *Head First* today! → ***Get Head First***

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